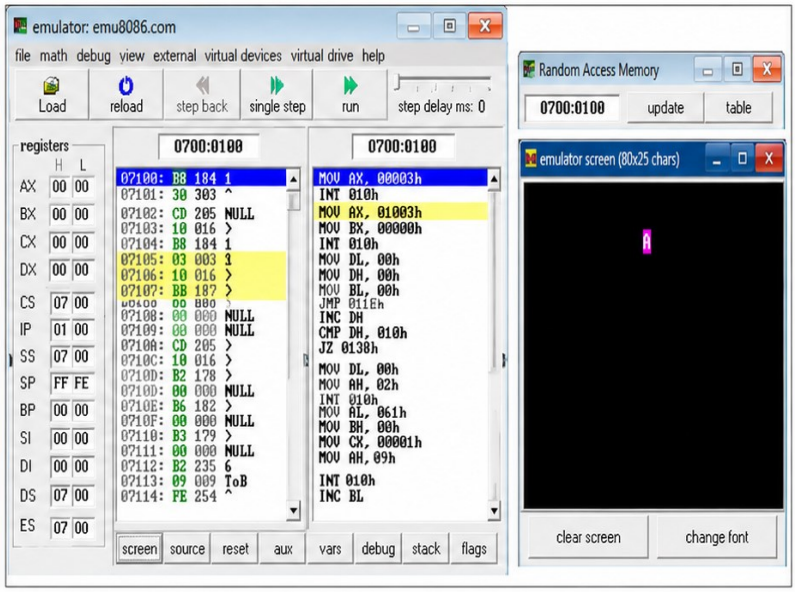
Lab 01

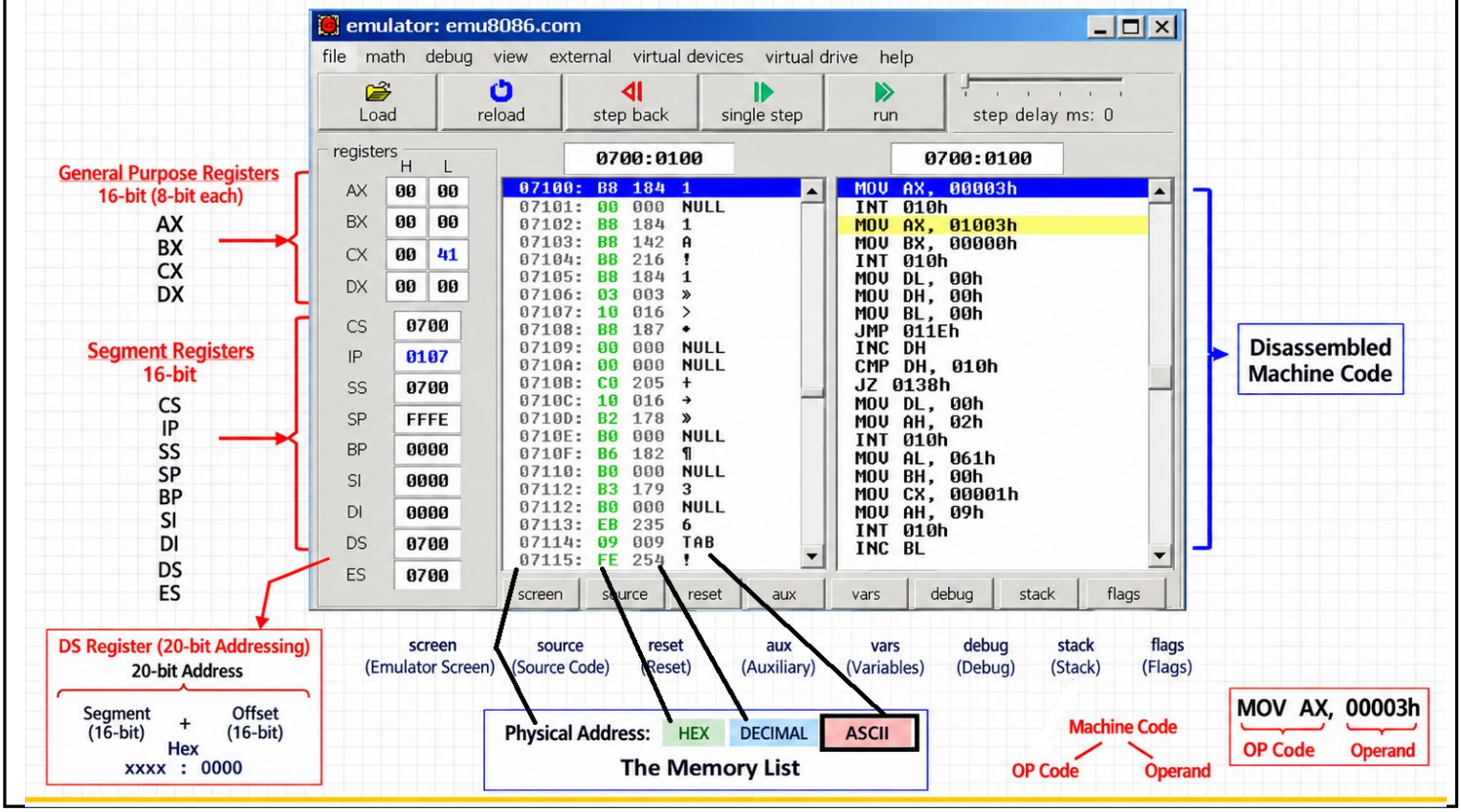
Introduction to Assembly Language

Object:

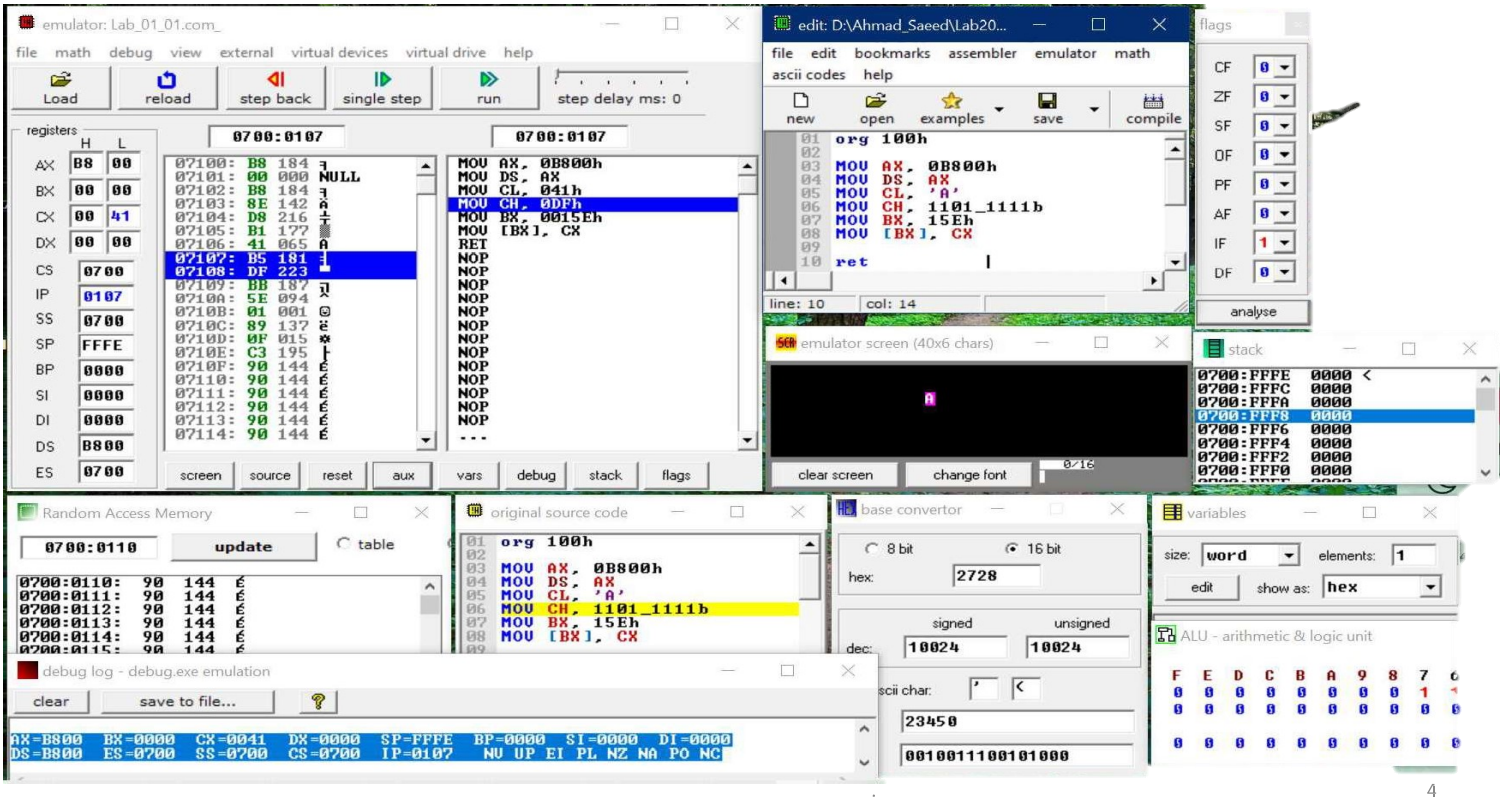
- Learn the emulator 8086 environment.
- Learn how to write and edit an assembly program.
- Learn how display some characters to emulator screen system.
- Learn how to send some information to a specific memory location.



The Environment of Emulator 8086

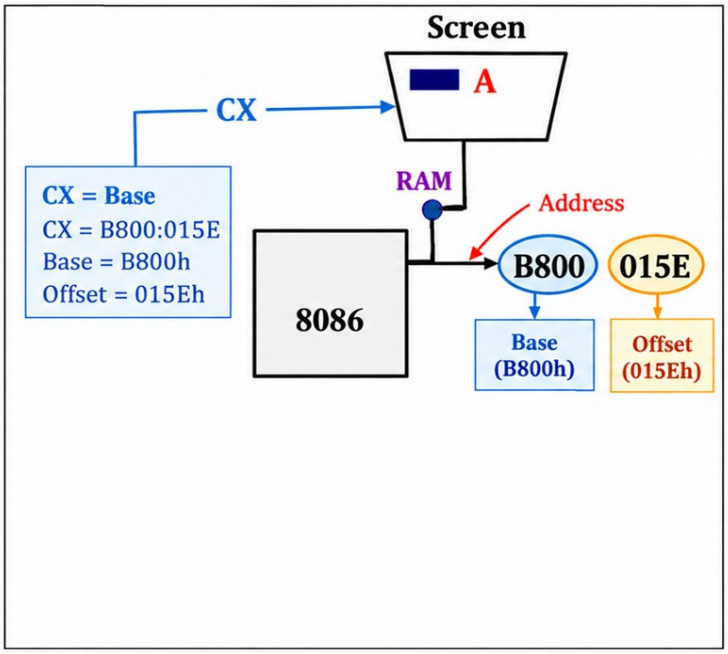


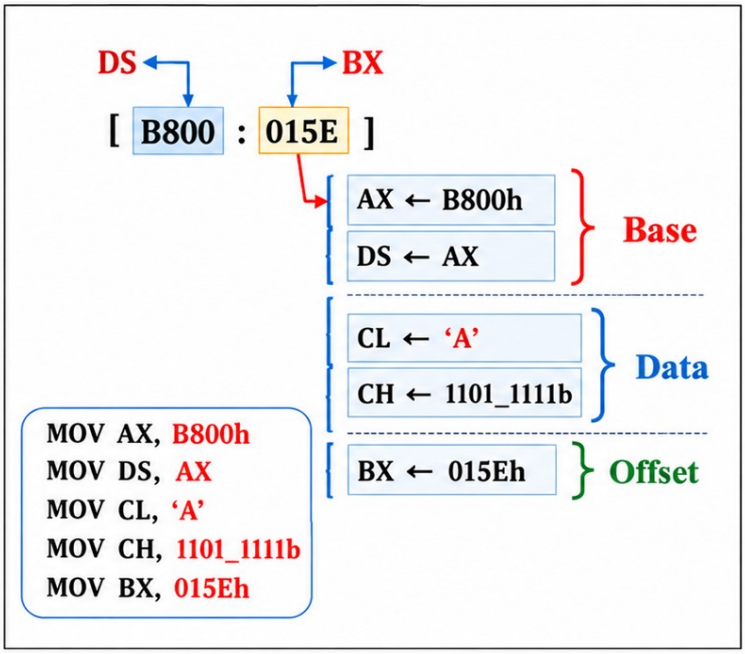
The Windows of Emulator 8086



Example 1

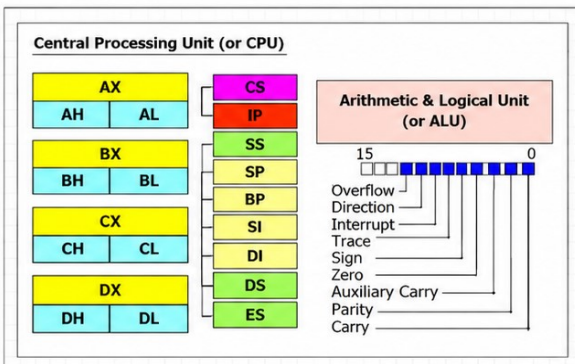
Write an assembly code to display a two characters which are top long block and ‘A’ in the emulator screen. The screen is related to the memory location B800:015E H. The top long block should be send as a binary and ‘A’ should be send as a character.

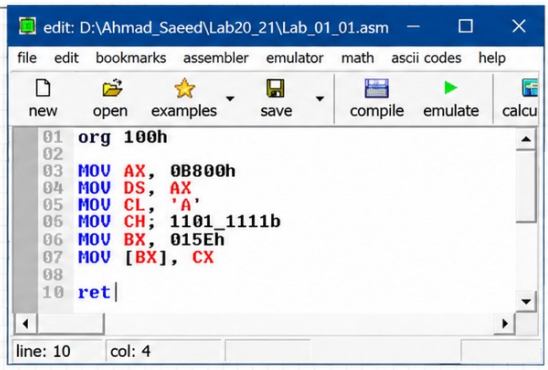


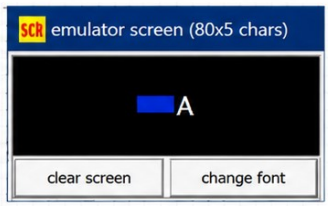


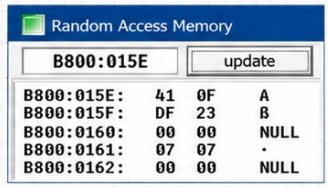
Example 1

Write an assembly code to display a two characters which are top long block and ‘A’ in the emulator screen. The screen is related to the memory location B800:015E H. The top long block should be send as a binary *and* ‘A’ should be send as a character.









registers

	H	L
AX	B8	00
BX	00	00
CX	00	00
DX	00	00

registers

	H	L
SI	0000	
DI	0000	
DS	B800	
ES	0700	

registers

	H	L
AX	B8	00
BX	00	00
CX	00	41
DX	00	00

registers

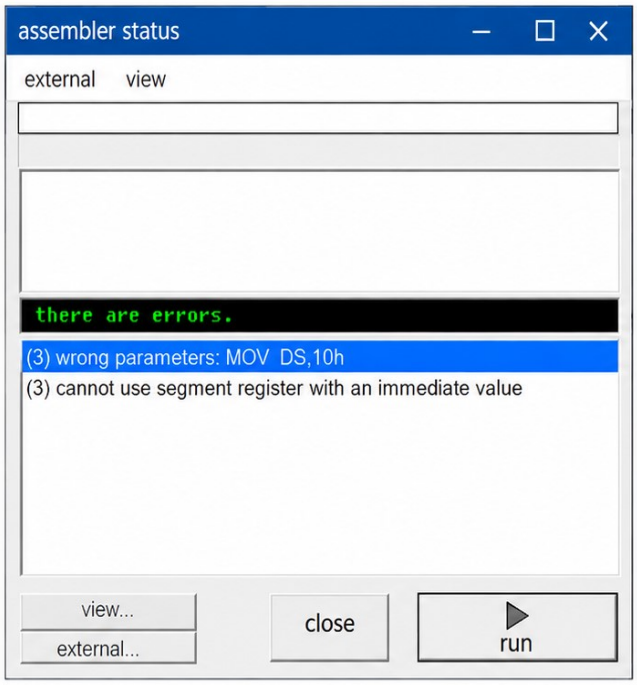
	H	L
AX	B8	00
BX	00	00
CX	0F	41
DX	00	00

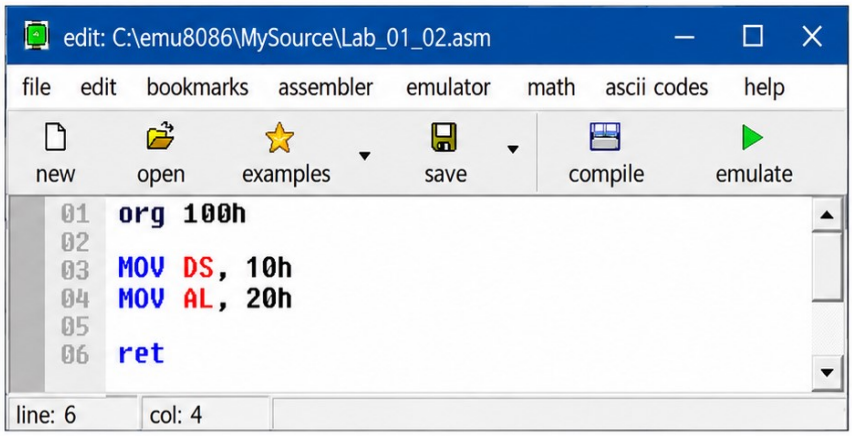
registers

	H	L
AX	B8	00
BX	01	5E
CX	0F	41
DX	00	00

Example 2

Try to load DS register directly with immediate data (10h), what error you will get? Why this happened as a hardware? Also try to load register AL with directly with immediate data (20h).





SEGMENT REGISTERS

- **CS** - points at the segment containing the current program.
- **DS** - generally points at segment where variables are defined.
- **ES** - extra segment register, it's up to a coder to define its usage.
- **SS** - points at the segment containing the stack.

